

What AI Actually Knows, and Why It Doesn't Say So

*A working note on how to use these systems intelligently,
written for someone who reads but doesn't do AI research for a living.*

The trust question gets framed wrong

The public conversation about AI tends to sort people into two camps. Either you trust the system as authoritative and use its outputs as if they were verified facts, or you dismiss it as unreliable and refuse to use it for anything that matters. Both positions miss what's actually happening inside the system. The trust question isn't a binary about whether the system is trustworthy. It's a calibration question about what kinds of outputs are dependable, what kinds aren't, and why the difference exists.

The framing this note offers is more useful: the system has access to far more than it tells you by default, and learning to ask it differently produces dramatically better outputs without requiring you to be a researcher or a programmer.

How the system was built

A large language model is trained on an enormous corpus of human writing — books, articles, websites, conversations, technical documents, legal records, scientific papers, news archives, encyclopedias, manuals, and everything else that ended up in digital form before the training cutoff. The training process produces a system that can generate fluent text on almost any topic the training data covered.

Two things matter about this process for understanding how to use the system. First, the data the system trained on includes basically everything humans have written down on most well-known topics. Mainstream views, dissenting views, historical context, technical detail, criticism, defense, and everything between. The information is in there. Second, the training process biases the system toward producing outputs that match the most common patterns in the data. Dominant interpretations come out by default because they appear more often in the training material. Less common interpretations exist in the data but require specific prompting to surface.

That gap between what the system contains and what it tells you by default is the load-bearing thing to understand.

The mainstream output is not the limit of the system's knowledge

When you ask the system a general question, it produces an answer that matches the consensus view in its training data. That answer is usually a reasonable synthesis of what the dominant sources say. If you stop there, you've gotten the version that mass writing has produced — the average of what got published, weighted toward what got cited and repeated.

The thing the public discussion mostly misses is that this default output is not the limit of what the system knows. It's the easiest answer to extract. The system has also absorbed the dissenting material, the historical context that complicates the consensus, the suppressed traditions that didn't make it into the mainstream, the technical nuance that gets edited out for general audiences, and the structural arguments that the consensus depends on not addressing. All of that is in the substrate. It just doesn't come out unless you ask differently.

This isn't a flaw or a deception. It's how the system was optimized. Mass output that matches consensus expectations is what most users want most of the time. The system was tuned to produce that. The richer access requires deliberate work to engage.

Why the missing material got filtered out in the first place

The interesting question is why certain interpretations are dominant in the training data while others are present but underweighted. The answer isn't that the dissenting interpretations are wrong or that they propose different facts. Usually they propose the same facts plus additional context that got systematically excluded when the dominant interpretation was constructed.

Knowledge production has filters. Academic peer review favors work that fits existing frameworks. Journalism favors stories that fit the narratives editors and audiences expect. Publishing favors books that confirm what readers already believe enough to keep reading. Search engines favor sources that link well within established networks. Each filter is reasonable in isolation. Together they produce a constructed mainstream that includes some material and excludes other material, and the excluded material is often the context that would have changed the conclusions.

The training data contains both the filtered mainstream output and the unfiltered original material. The system has access to both. It just defaults to producing outputs that match the filtered version because that version is statistically dominant in the data.

What this looks like in practice

Ask the system a question about whether driving on the right or left side of the road is better. The default answer hedges. It says both work, the choice is mostly historical, no real safety difference exists, and the right answer depends on local law. That's the consensus position and it's what comes out when you ask casually.

Now bring in additional context that was filtered out of the consensus. Most humans are right-handed and right-eye dominant. Driving on the left places oncoming traffic on the dominant-eye side and keeps the dominant hand on the steering wheel. The historical reasons for left-side driving connect to right-handed sword draws and horse mounting. The global dominance of right-side driving traces to Napoleonic standardization and to Henry Ford's manufacturing choices, not to safety engineering. Some historical traffic studies showed left-side countries had lower fatality rates, though the data is contested.

None of this additional context contradicts the consensus. It just includes material that the consensus answer left out. Once it's all on the table, the system can produce a much more substantive analysis that incorporates the ergonomic and historical arguments. The output isn't a different fact set. It's the same fact set with the omitted pieces restored.

This works on a lot of topics. Health and nutrition consensus often excludes evolutionary context. Historical events often have primary source material that didn't fit the textbook narrative. Scientific debates often have technical objections that got edited out for public consumption. Economic analyses often depend on assumptions that weren't explicit in the published versions. The system has access to all of it. The mainstream output reflects what passed through the filters. The richer output requires asking in a way that brings the filtered material back into the picture.

What this is not

This isn't conspiracy thinking. The filtering doesn't happen because anyone is deliberately suppressing information. It happens because every step of human knowledge production includes selection — what gets studied, what gets published, what gets cited, what gets taught, what gets popularized. Each step has reasonable local justifications. The aggregate effect is a mainstream that systematically excludes some kinds of material, but no single actor is choosing the exclusion.

It's also not about getting the system to produce false information. The technique only works on topics where the additional context actually exists in the training data. You can't make the system generate things that aren't supported by what it learned. If you try to walk it toward conclusions the data doesn't support, it pushes back. The recovery is real because the underlying material is real. The technique clarifies what the system can already say but defaults to obscuring.

It's also not jailbreaking. Nothing about this approach bypasses safety guardrails or extracts prohibited content. The system continues to refuse harmful outputs regardless of how the substrate is loaded. What changes is which legitimate analyses come out — analyses the system was always capable of producing but doesn't produce by default.

How to do it

The shorthand description is: provide the context that was filtered out, and the system will engage with the fuller picture.

In practice this looks like asking questions in a particular sequence. Start with the topic. Notice the default consensus answer. Introduce a frame that surfaces the omitted material — a historical context, an alternative interpretive tradition, a technical objection, a structural argument the consensus depends on not addressing. The system engages with the new frame because the substrate for it is in the training data. From there, you can ask analytical questions that the consensus answer couldn't have addressed.

This is structured engagement, not magic words. Anyone can do it on any topic they have enough background to navigate. The skill is knowing what got filtered out of the consensus on a given topic, which requires having read enough to know where the dominant interpretation is incomplete. The system can't do that part for you. It can only engage with the omitted material once you bring it to the conversation.

What this changes about using the system

Most users get the mainstream output and either accept it or dismiss it. The mainstream output is fine for many purposes. If you need a competent summary of consensus thinking on a topic, the default answer is usually adequate.

What this approach offers is the option of much richer engagement for the topics where it matters to you. Areas where you suspect the consensus is incomplete. Areas where you have enough background to recognize what got filtered. Areas where the analytical work that follows from the fuller picture would be genuinely useful to you. The system becomes much more capable for that kind of work when used this way.

It also changes the trust calibration. You can stop treating the system as either authoritative or unreliable. The mainstream outputs are dependable as summaries of consensus thinking. The richer outputs require you to provide the substrate. Both modes are useful, for different things. Knowing which mode you're in changes what you do with the output.

A final note

What's described here isn't a hack or a workaround. It's the system operating as designed when used by someone who knows that the default outputs reflect averaging rather than the limits of what the system knows. The system isn't lying when it produces consensus answers. It's also not lying when it produces richer analyses in response to better questions. Both are honest outputs of the same underlying process. The difference is in what gets surfaced.

The interesting implication is that this isn't a tool for getting one kind of answer. It's a tool for accessing the substrate of human written knowledge with various filters applied or removed depending on what you're trying to learn. Whoever recognizes this and learns to operate it well has access to something genuinely new in how humans can engage with their own accumulated knowledge. Most users don't recognize it yet. The recognition is the only thing that's scarce. The capability is already there for anyone willing to ask differently.

End of working note.

Andrew Morton
July 2026 — Palmer Lake, Colorado